NYCU Introduction to Machine Learning, Homework 4

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Part. 1, Kaggle (70% [50% comes from the competition]):

(20%) Introduction of your idea, methods, and key to success

Data Processing & Preprocessing

Dataset description: Binary image classification

Class 0: Real (human-generated)，Class 1: AI-generated

Image Preprocessing:

Resize all images to 380 × 380 to match EfficientNet-B4 input size

Convert images to RGB format and Normalize mean and standard deviation

Data Augmentation(Training):

Random horizontal flipping

Mild rotations for robustness

```
train_transform = transforms.Compose([
    transforms.Resize((380, 380)),
    transforms.RandomHorizontalFlip(),
    transforms.ToTensor(),
    transforms.Normalize(
        mean=[0.485, 0.456, 0.406],
        std=[0.229, 0.224, 0.225]
    ),
])
```

Semi-supervised learning with unlabeled data

Used model predictions to generate pseudo-labels for unlabeled images

Only samples with high confidence (≥ 0.99) were selected

Pseudo-labeled samples were combined with original training data

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Model Architecture

Backbone selection:

EfficientNet-B4 (pretrained on ImageNet)，Chosen for its strong ability to capture fine-grained textures and frequency patterns

Architecture modification:

Replaced the final classifier layer with a 2-class linear layer

Output: {Real, AI-generated}

Fine-tuning strategy:

Freeze most backbone parameters

Unfreeze classifier (and optionally last block) for fine-tuning

Performance improvement:

EfficientNet-B4 significantly outperformed ResNet-based models

Higher public F1 score due to better feature representation

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Training Strategy & Performance Boost

Training setup:

- Loss function: Cross-Entropy Loss
- Optimizer: Adam / AdamW
- Batch size: adjusted for GPU memory
- Training on GPU

Semi-supervised training:

- Generated pseudo-labels from unlabeled data
- Only high-confidence predictions (≥ 0.99) were used
- Combined labeled and pseudo-labeled data for final training

Test-Time Augmentation:

- Applied horizontal flip (and rotation) during inference
- Averaged predictions across augmented versions

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Appendix

108	112550068		0.98691	2	1d
109	112550062		0.98545	2	25m
110	112705007		0.98446	7	1m
Your Best Entry! Your most recent submission scored 0.85190, which is not an improvement of your previous score. Keep trying!					
111	111550176		0.98137	10	37m
112	112550177		0.97132	4	9h

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Part. 2, Questions (30%):

1. (10%) Based on the "SVMs vs. Logistic regression" lecture slide, explain which kinds of training data points mainly determine the classifier learned by SVM and which types of points influence Logistic Regression, and briefly justify your answer by referring to the shapes of hinge loss and logistic loss.

   In SVM, the learned classifier is mainly determined by the training points that lie on or inside the margin(support vectors). This is because the hinge loss is zero for correctly classified points that satisfy $yf(x) \geq 1$, so points far away from the decision boundary do not affect the optimization.

Logistic Regression is influenced by almost all training data points. The logistic loss is pos itive for $yf(x)$, even for correctly classified points far from the decision boundary. As a re sult, every data point contributes to the parameter updates.
This difference arises from the shapes of the loss functions: hinge loss becomes flat beyon d the margin, while logistic loss decreases smoothly but never becomes zero.

2. (15%) For an SVM with a Linear Kernel, determine whether to use the Primal or Dual For m for the datasets below. Justify your choice based on: (a) Optimization variables & comp utational complexity, (b) Memory requirements (specifically the size of the Gram Matrix), (c) Prediction cost.
   - Dataset A: N=100, M=20,000
   - Dataset B: N=1,000,000, M=20

For Dataset A, the dual form is preferred.
(A) In the primal form, the number of optimization variables is M(=20000), which is very l arge. In contrast, the dual form only requires optimizing N(=100) variables.
(B) The Gram matrix has size 100×100, which is small and memory-efficient.
(C) Since the number of support vectors is limited, the prediction cost in the dual form is a cceptable.

For Dataset B, the primal form is more suitable.
(A) The primal optimization only involves M(=20) variables, while the dual form would re quire optimizing N(=1000000) variables, which is computationally expensive.
(B) The Gram matrix in the dual form would be very large, but in the primal form, the me mory will be positively relative to M(=20), which is small and effeicient.
(C) The primal form also allows very efficient prediction, since it only requires computing an inner product between the weight vector and the input feature vector.

3. (5%) To train a neural network, what do we need to optimize it? (How do we know the net work is good or not?) Also, what algorithm can we use to optimize the neural network? (th e most basic one).

To train a neural network, we need to optimize a loss function (MSE or cross entropy), which measures the difference between the network' s predictions and the true labels. The value of the loss function indicates how good or bad the network is, with a smaller loss implying better performance.

The most basic algorithm used to optimize a neural network is gradient descent. Gradients of the loss with respect to the network parameters are computed using backpropagation, and the parameters are updated iteratively to minimize the loss.